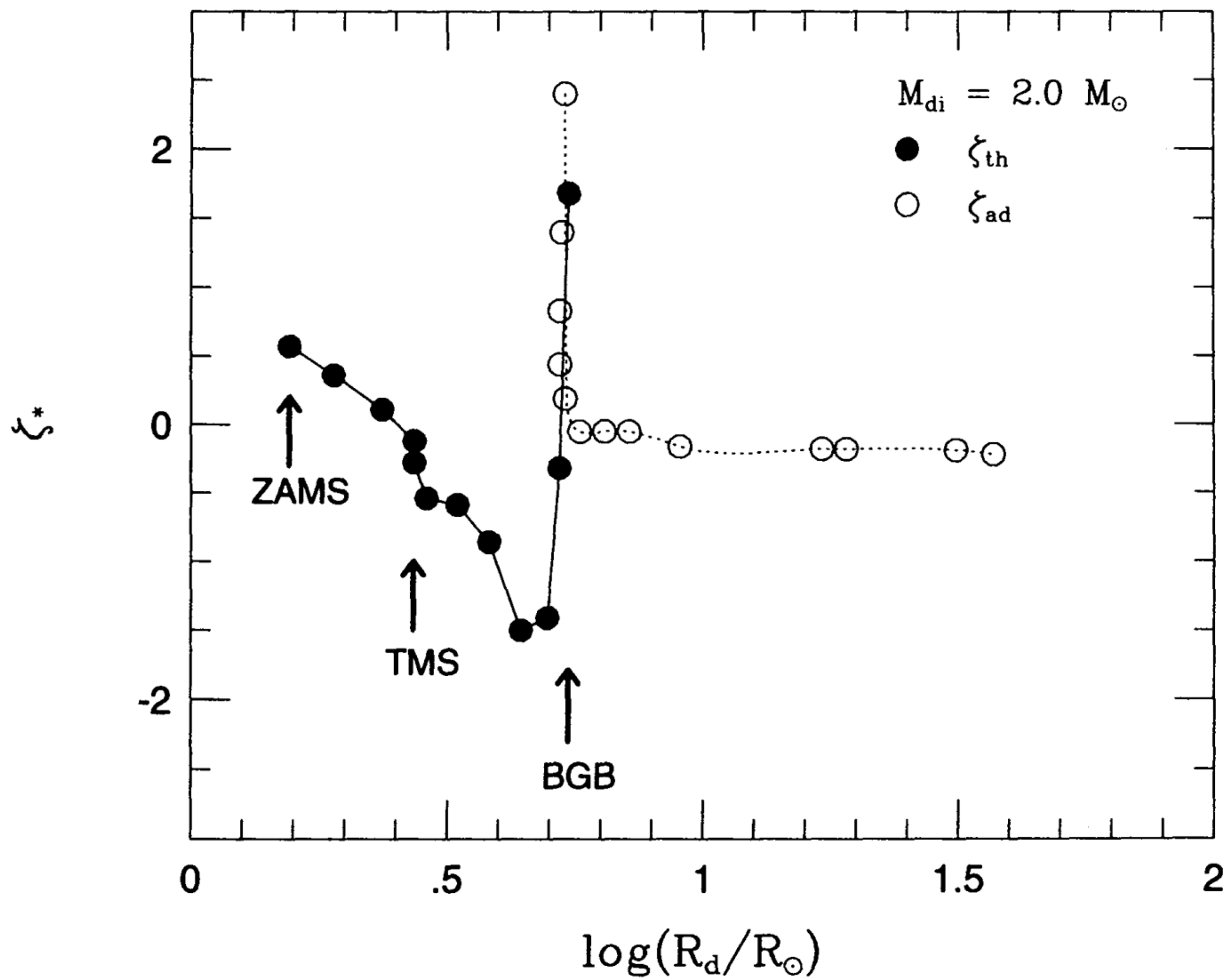


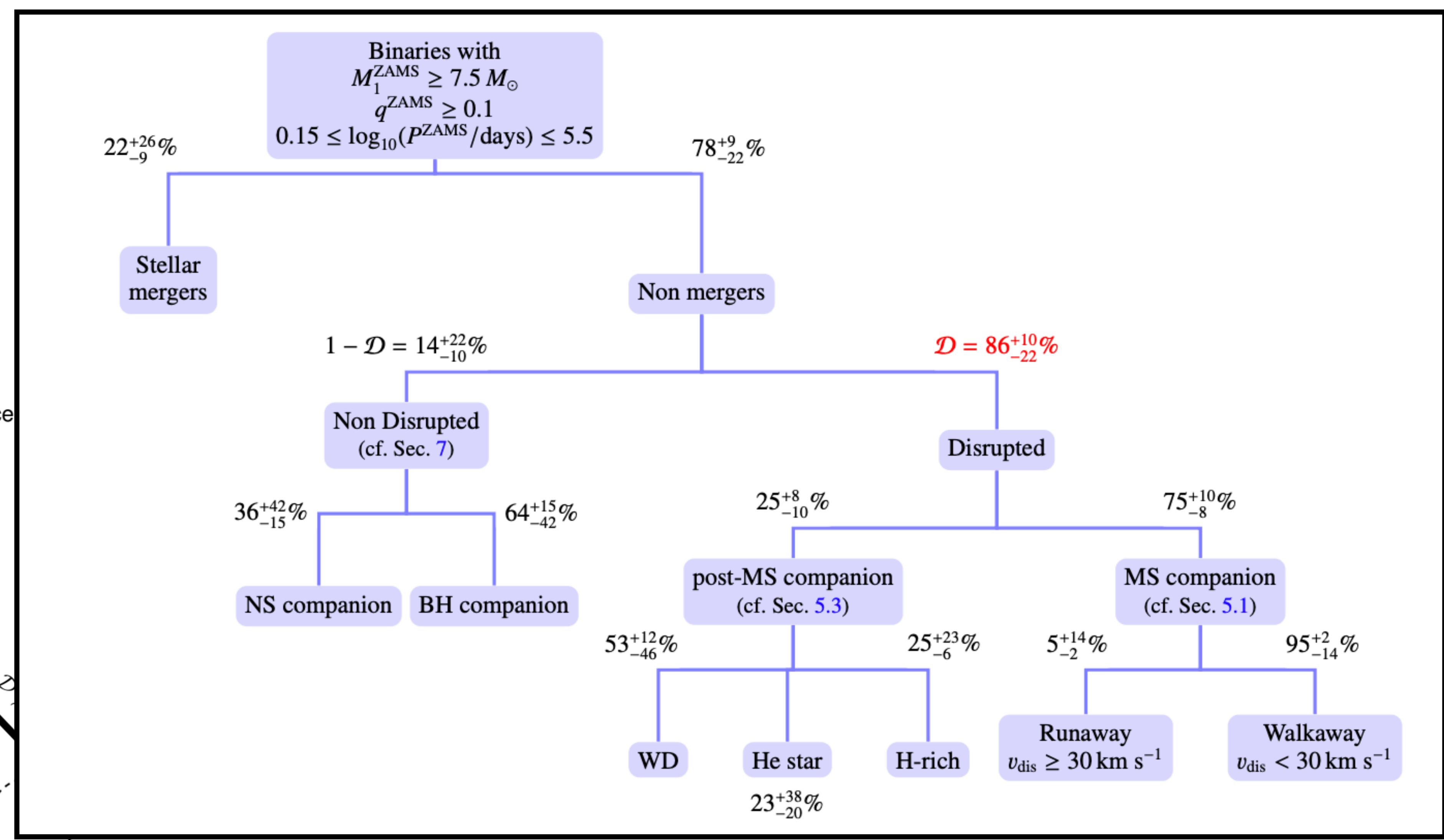
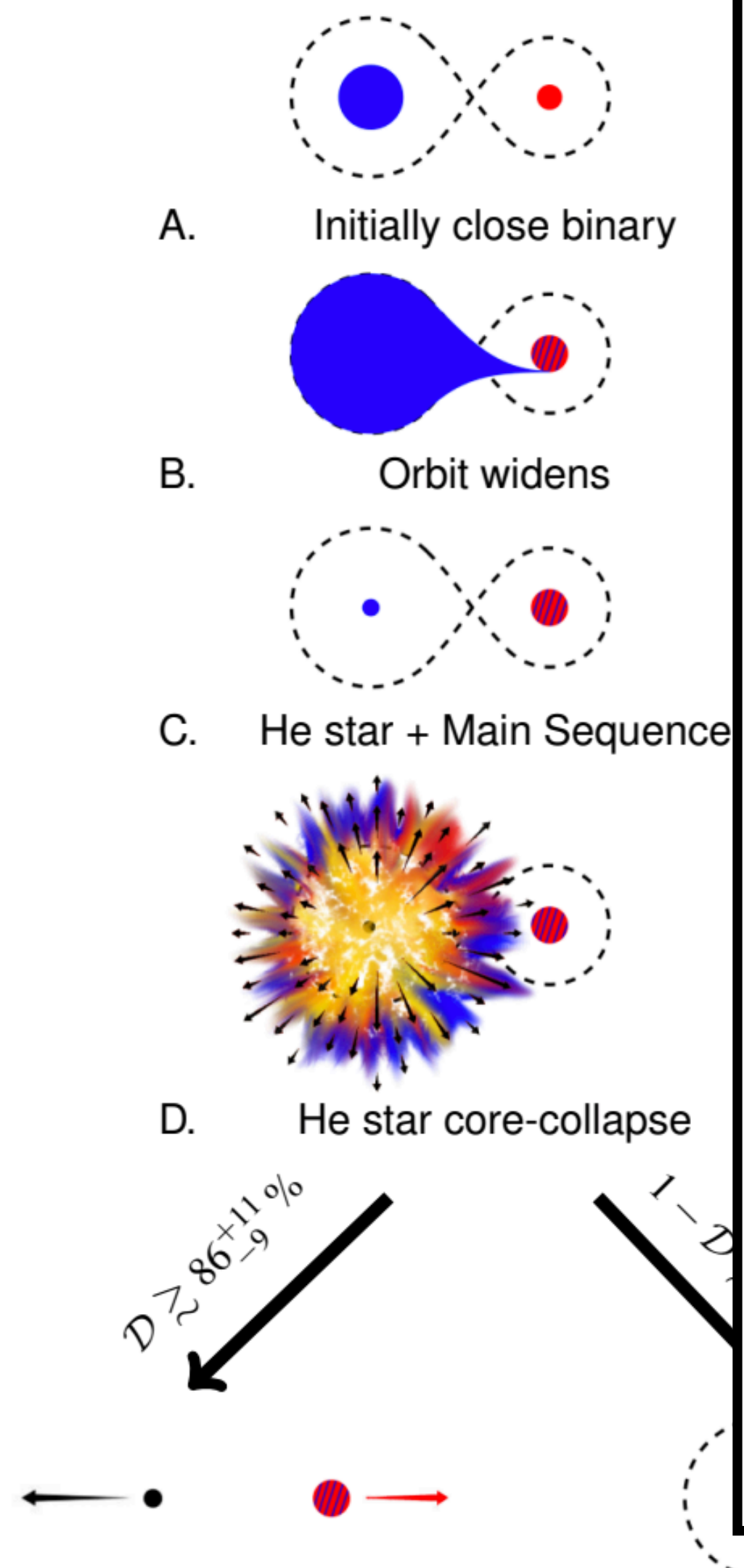
Quick announcements:

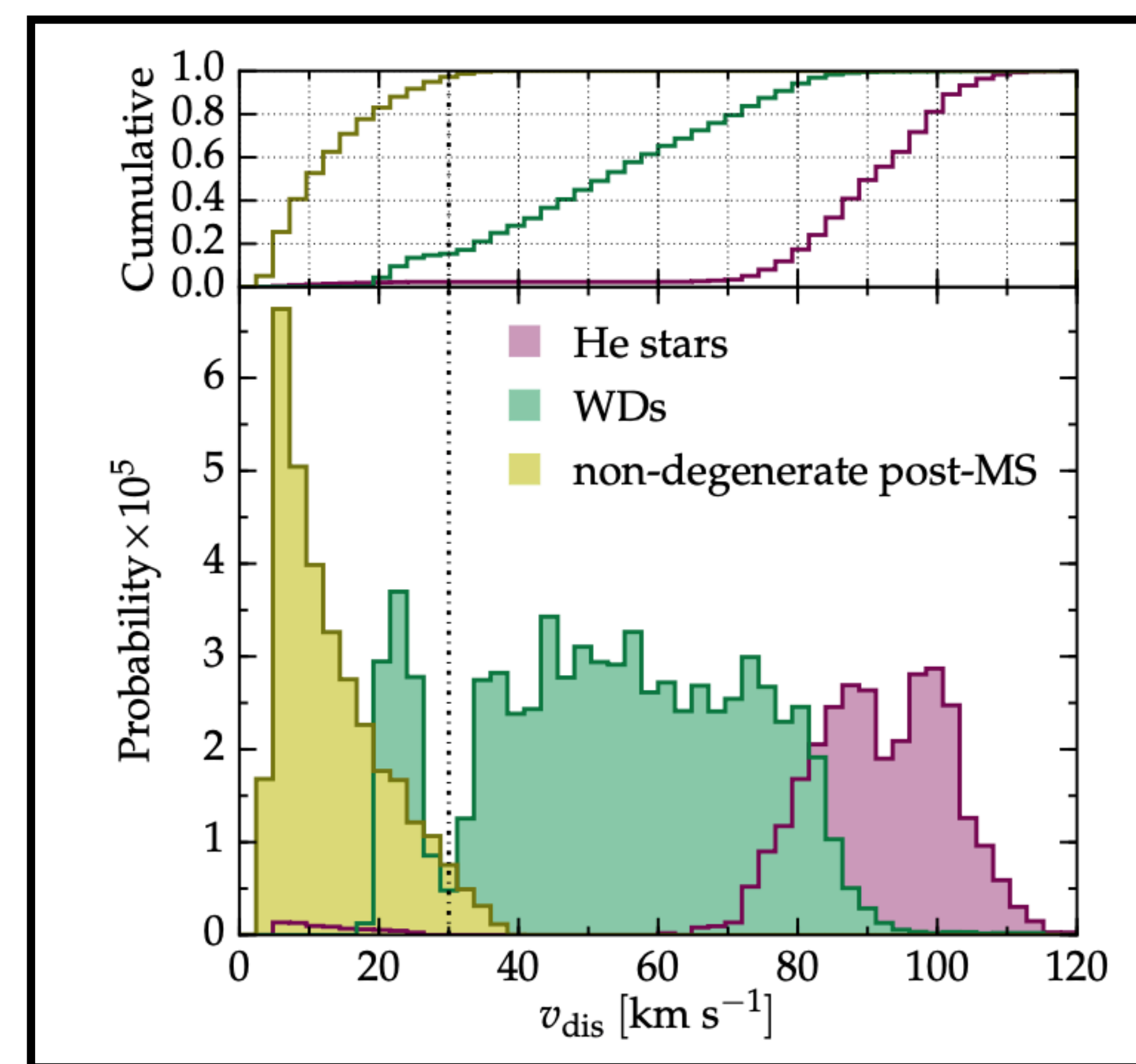
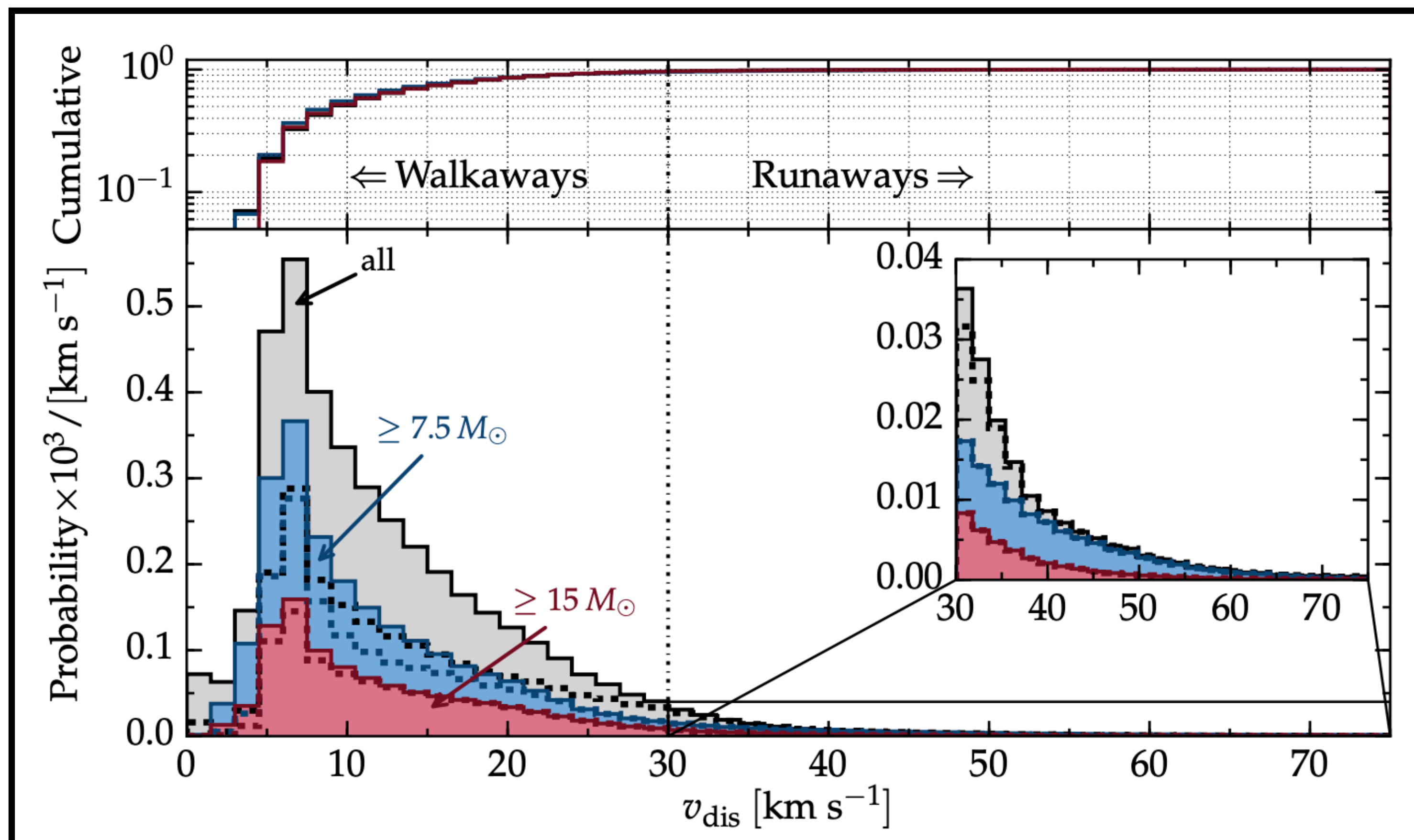
- HW 5 is due before class on Thursday
 - Take a look at the Canvas announcement for tips on working with the MIST data
- No office hours today due to the Departmental Review
 - If you need to meet, send me an email and we can find a time!
- You should have a project for the last half of the semester by now
 - If you don't yet have one with a new partner, you need to send me your plans by the end of class today.
- I have to take off at 11:30 for the departmental review
 - Project work time will start then and Brendan O'Connor will be hanging out in the room with y'all



$$\zeta_L = \frac{d \ln R_L}{d \ln M_d} = \frac{4}{3} (2q - 1) - (1 + q)F ,$$

$$F \equiv \frac{0.4q + [(1/3)q^{2/3}]/(1 + q^{1/3})}{0.6q + q^{1/3} \ln(1 + q^{1/3})} .$$



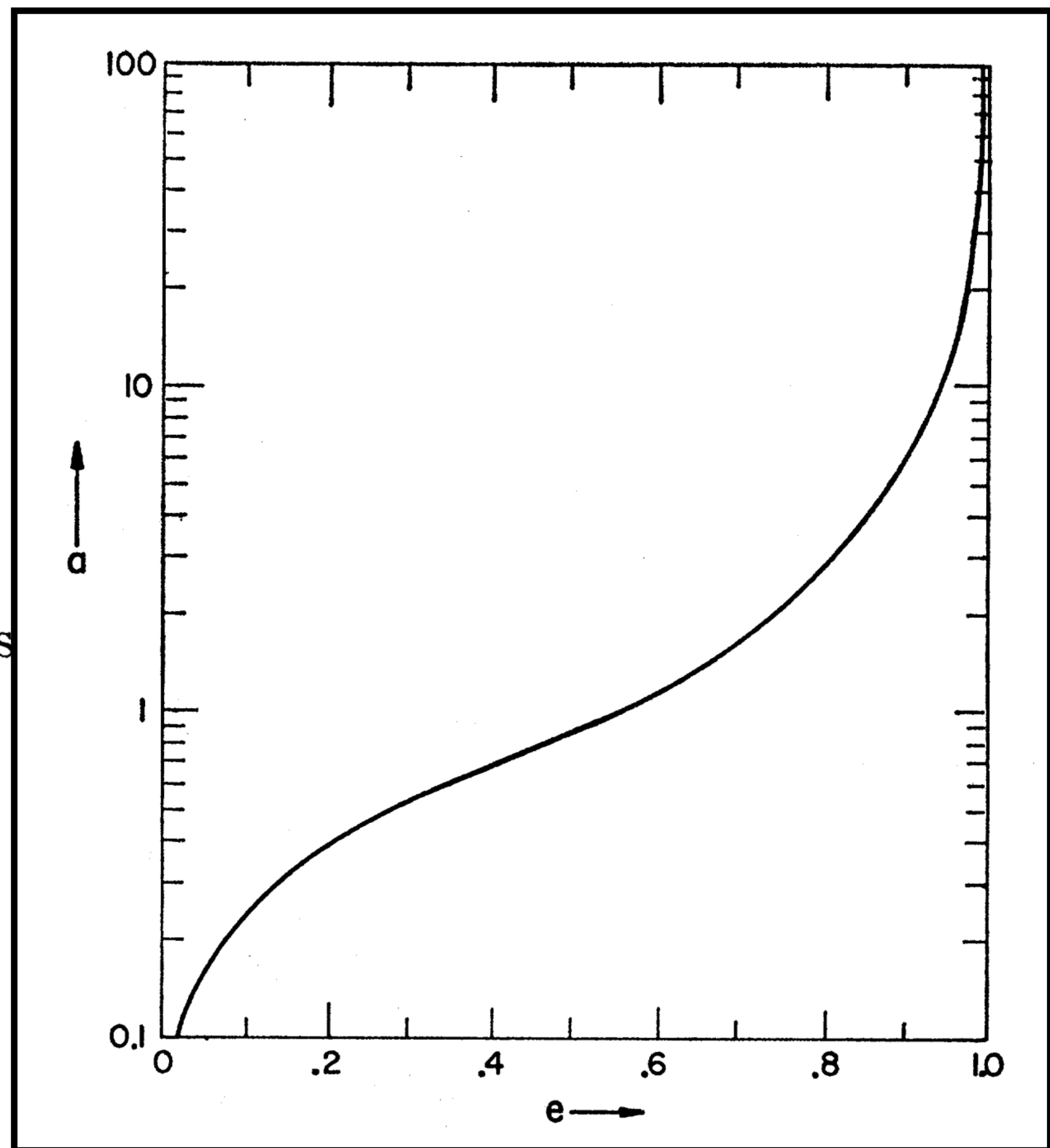


$$\left\langle \frac{dE}{dt} \right\rangle = -\frac{32}{5} \frac{G^4 m_1^2 m_2^2 (m_1 + m_2)}{c^5 a^5 (1 - e^2)^{7/2}} \left(1 + \frac{73}{24} e^2 + \frac{37}{96} e^4 \right). \quad (5.4)$$

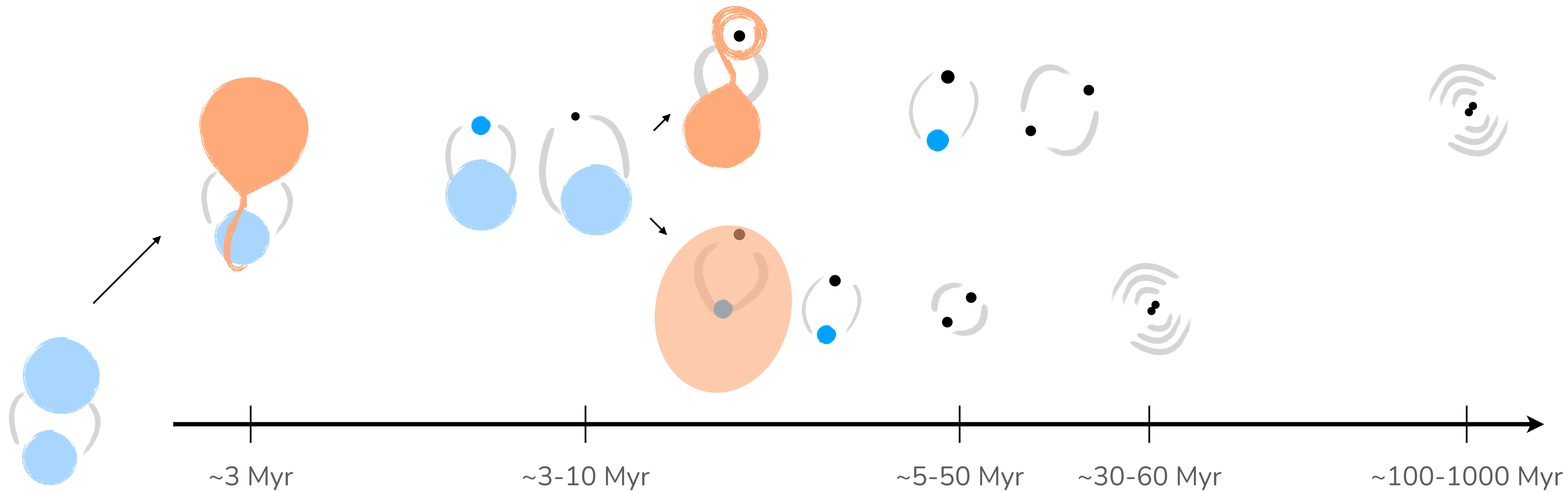
$$\left\langle \frac{dL}{dt} \right\rangle = -\frac{32}{5} \frac{G^{7/2} m_1^2 m_2^2 (m_1 + m_2)^{1/2}}{c^5 a^{7/2} (1 - e^2)^2} \left(1 + \frac{7}{8} e^2 \right). \quad (5.5)$$

$$\left\langle \frac{da}{de} \right\rangle = \frac{12 a [1 + (73/24)e^2 + (37/96)e^4]}{19 e (1 - e^2) [1 + (121/304)e^2]} \quad \text{this}$$

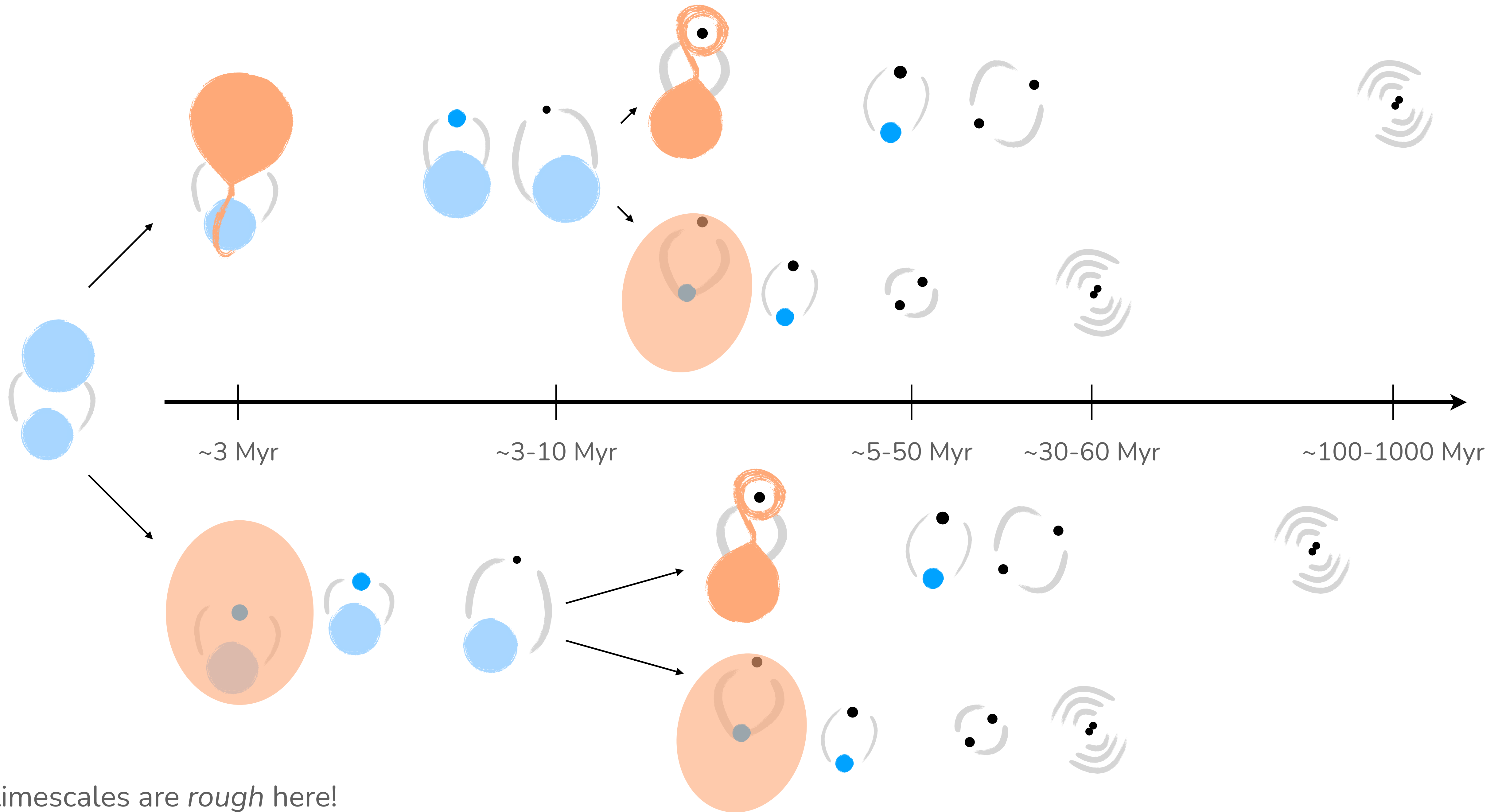
$$a(e) = \frac{c_0 e^{12/19}}{(1 - e^2)} \left[1 + \frac{121}{304} e^2 \right]^{870/2299},$$

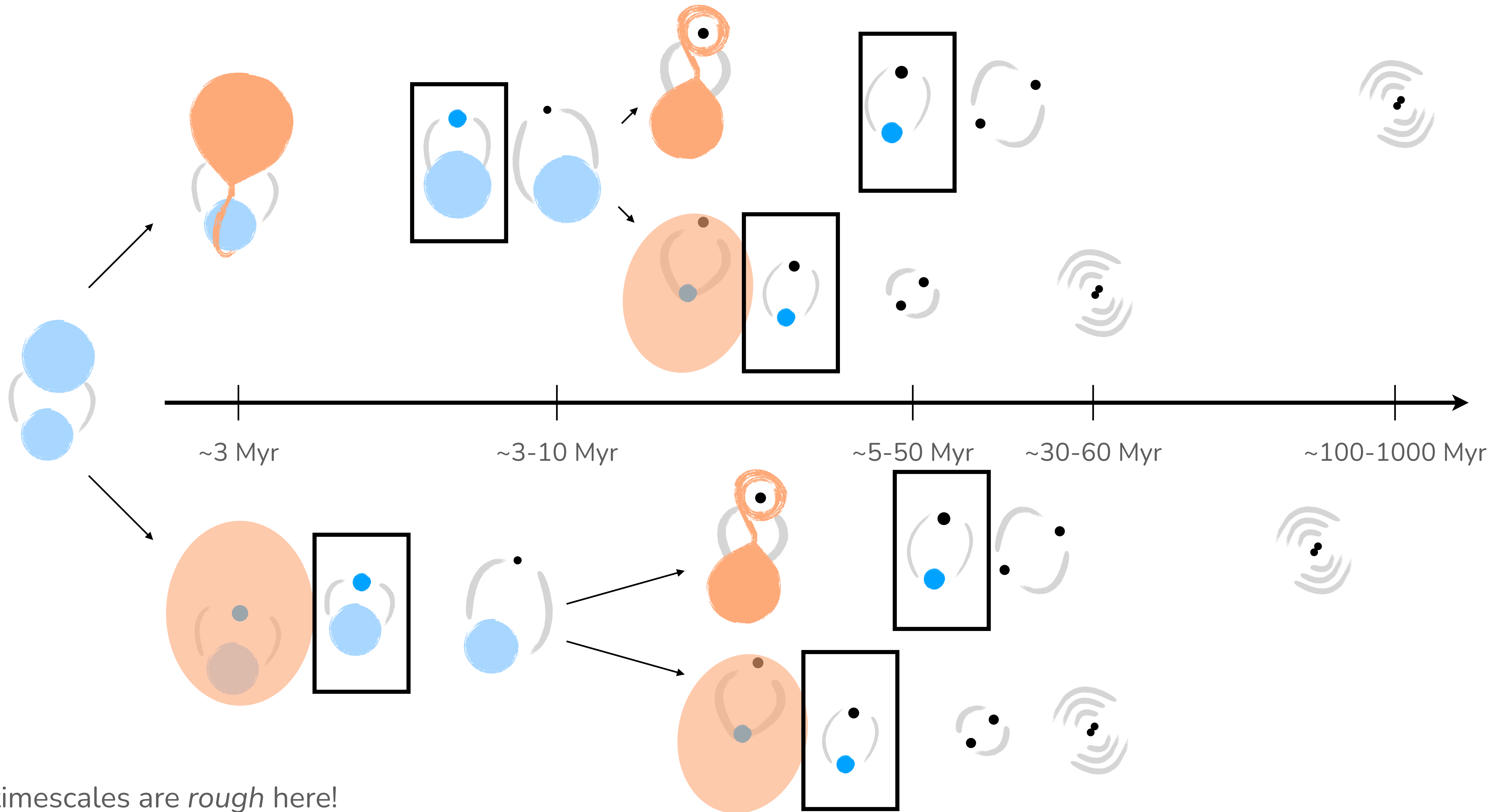


ion of

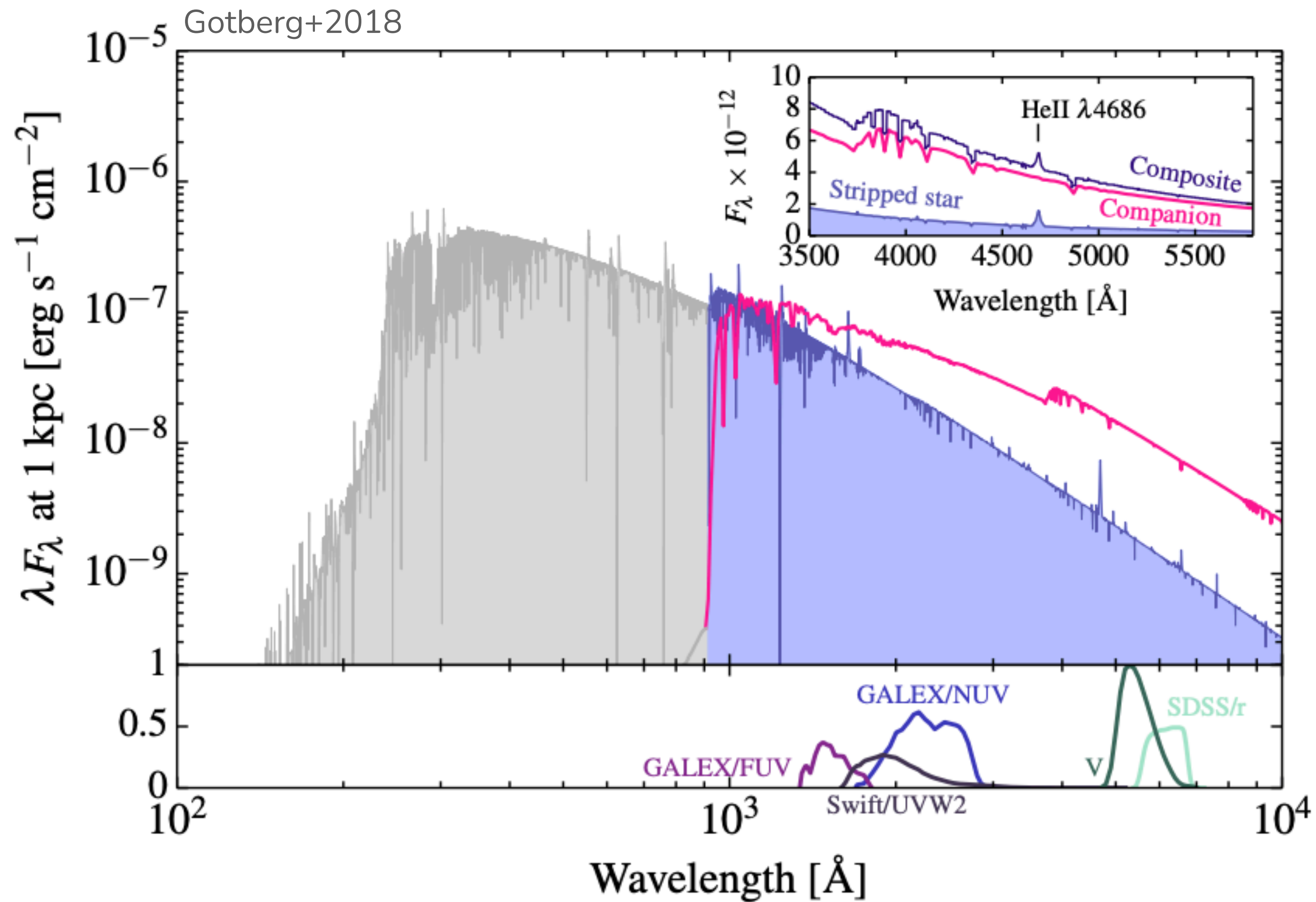


*timescales are *rough* here!

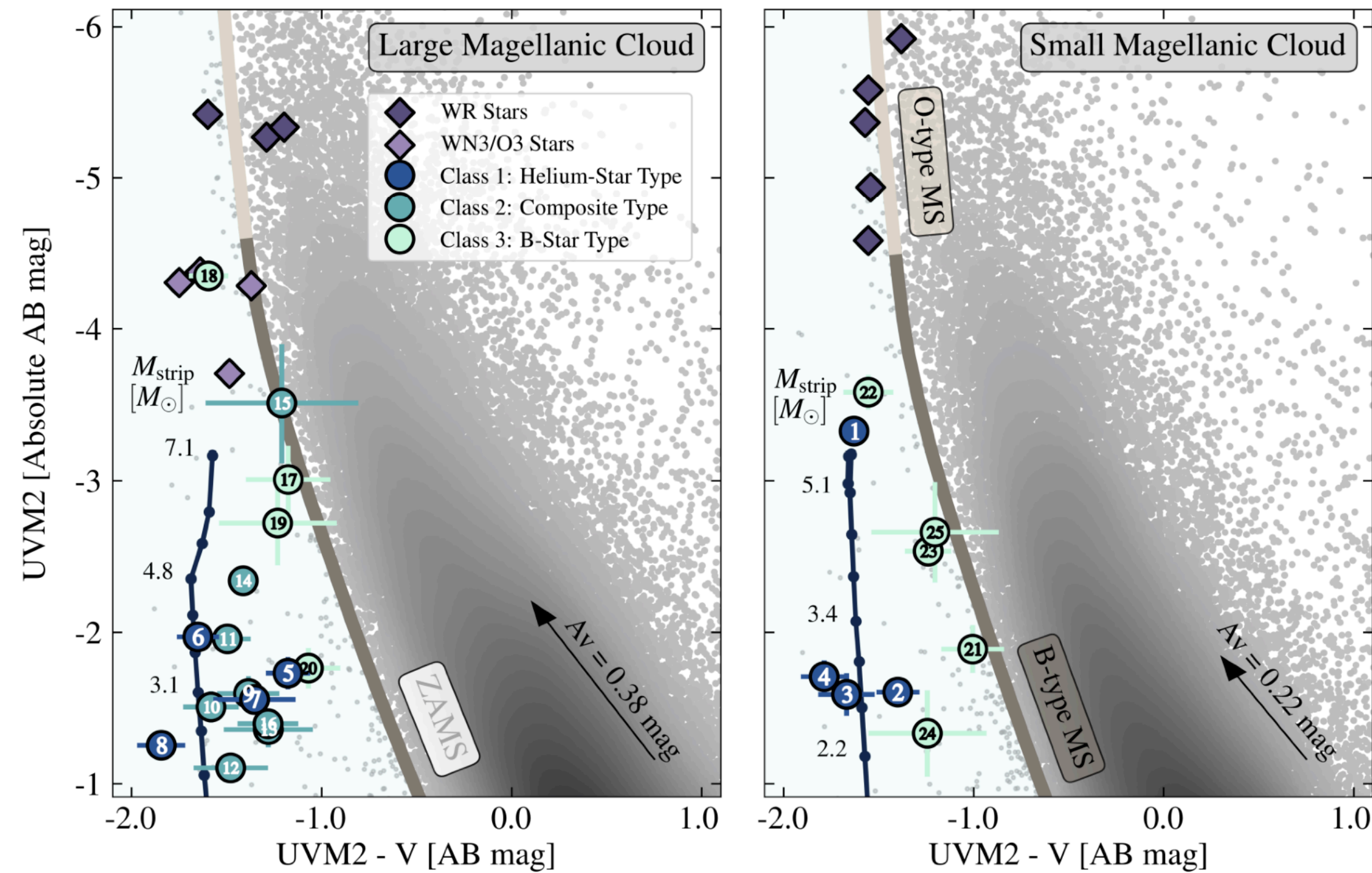




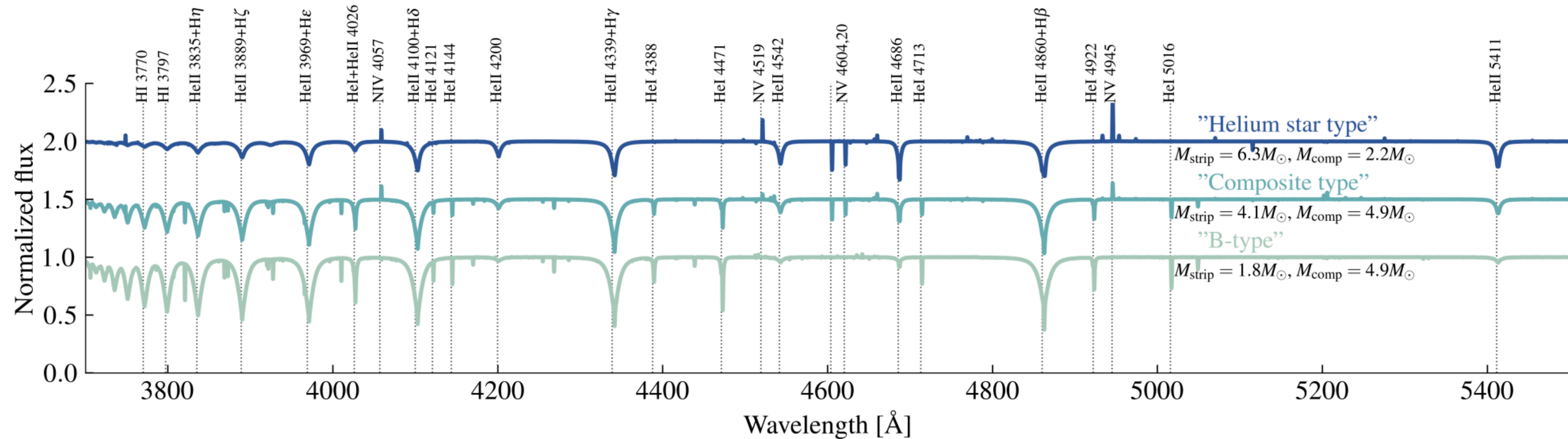
*timescales are *rough* here!

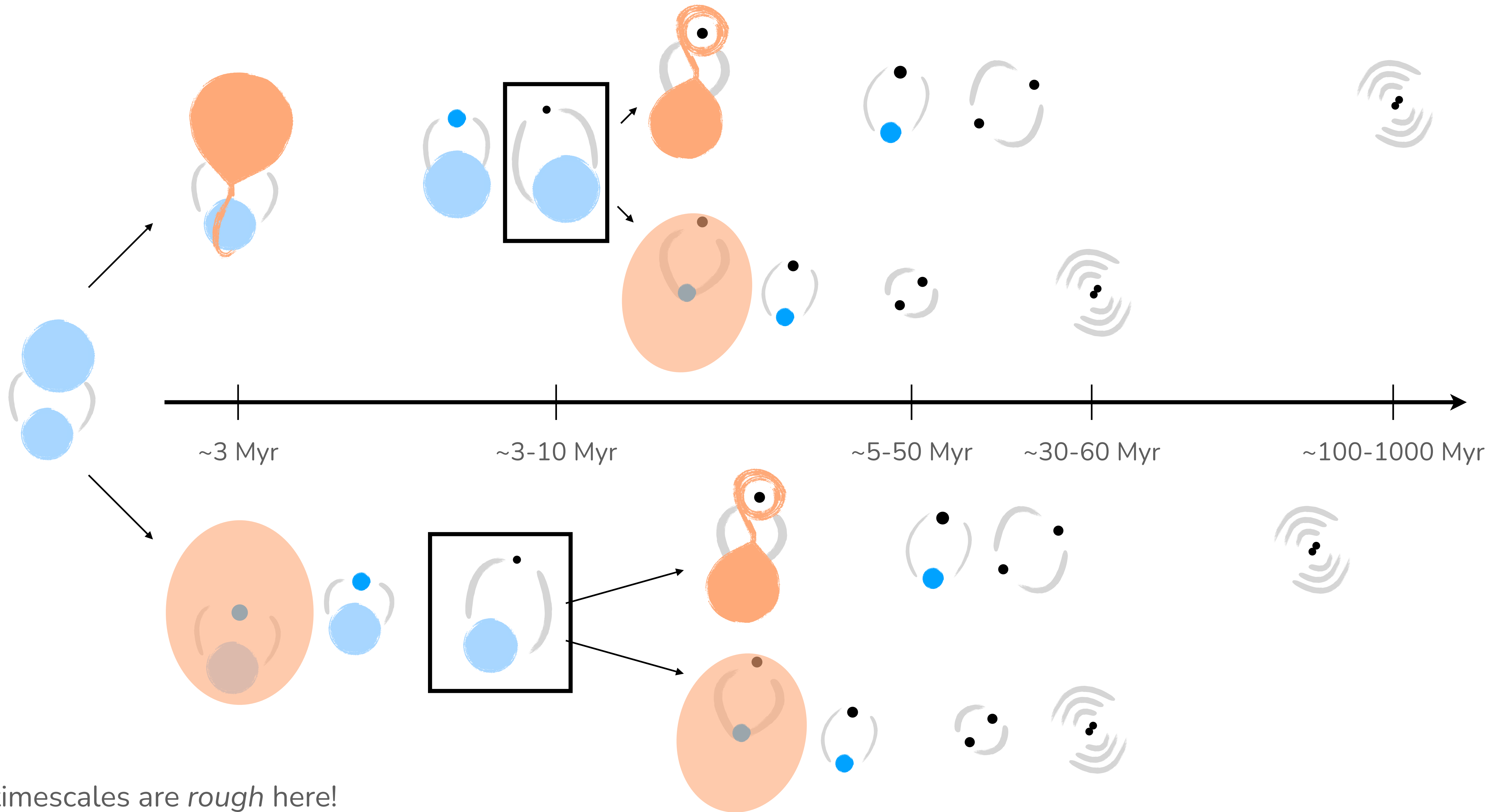


Most
instruments
can't see
stripped
stars at all!



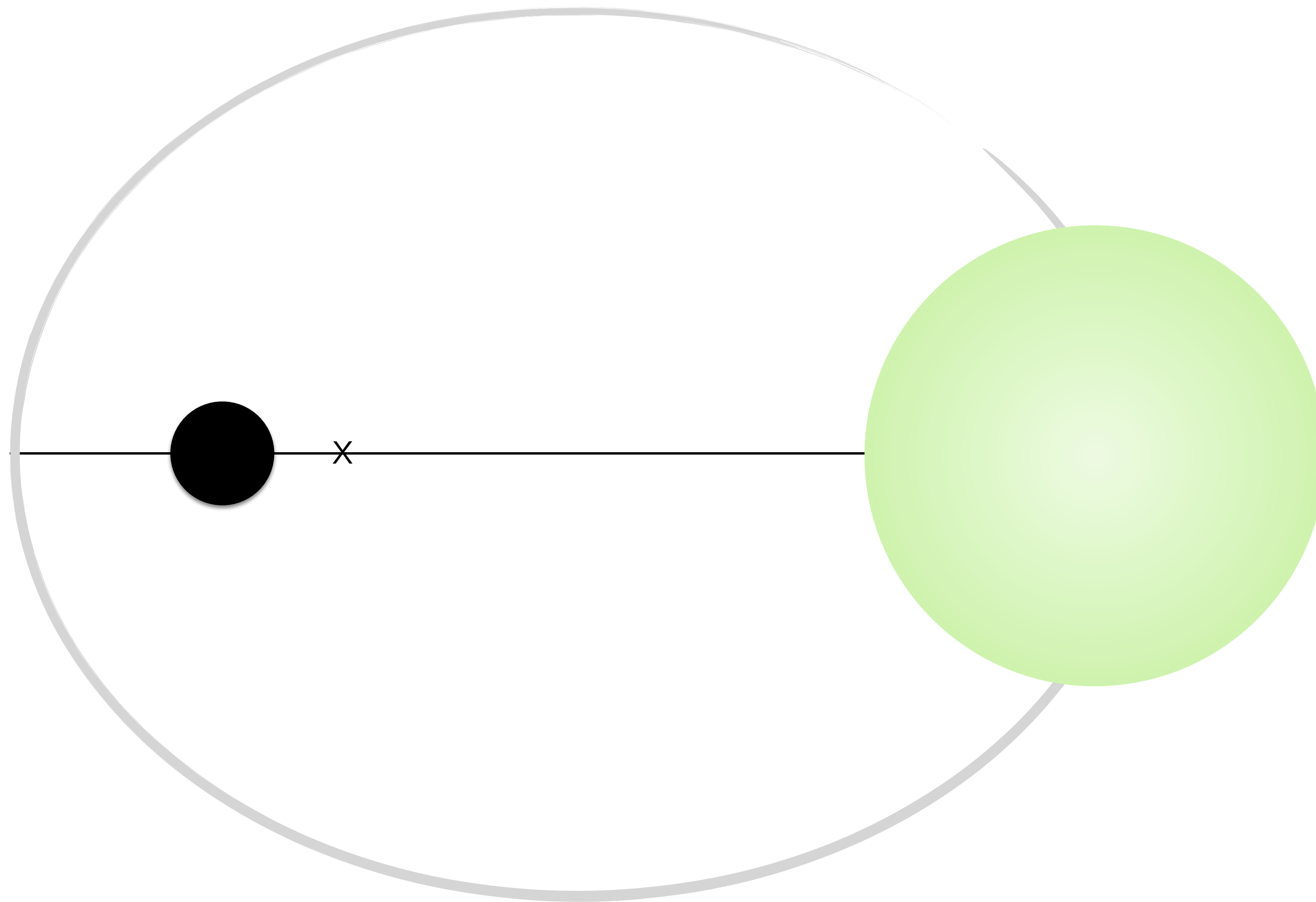
You have to look in
UV to find them —
but they are there!





*timescales are *rough* here!

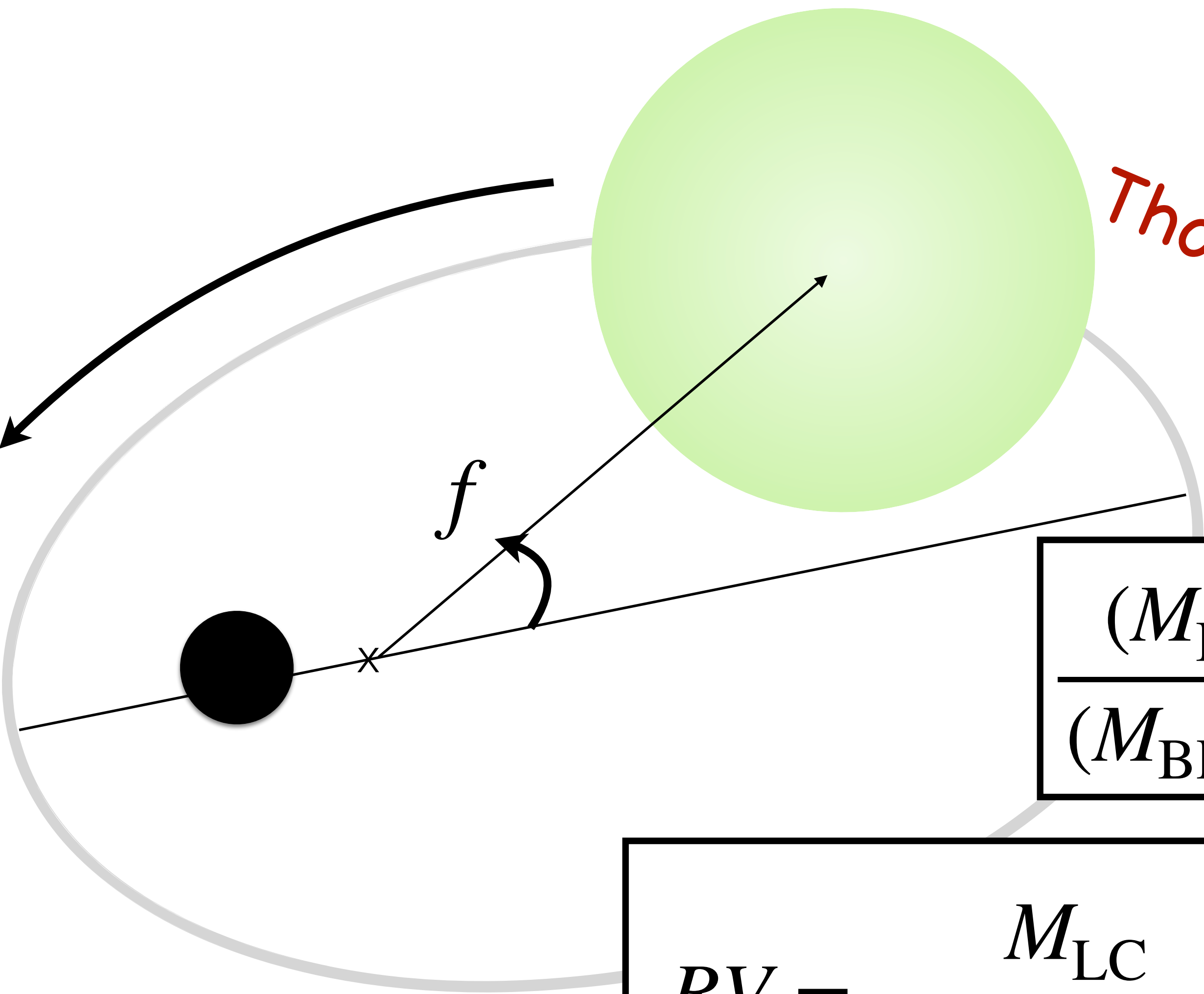
dormant black hole binaries



radial-velocity black hole binaries

*First predicted
by Trimble &
Thorne in the 60s!*

e.g. Andrews, [KB+2019](#)


$$\frac{(M_{\text{BH}} \sin I)^3}{(M_{\text{BH}} + M_{\text{LC}})^2} = \frac{K^3}{G} \left(\frac{P_{\text{orb}}}{2\pi} \right) (1 - e^2)^{3/2}$$

$$RV = \frac{M_{\text{LC}}}{M_{\text{BH}} + M_{\text{LC}}} \frac{2\pi}{P_{\text{orb}}} \frac{a}{1 - e^2} (\cos f \sin I + e \sin I)$$

2 BH candidates in NGC 3201

Giesers+18,19

$$M_{\text{BH}} = 4.53 \pm 0.21 M_{\odot}$$

$$M_{\star} = 0.81 \pm 0.05 M_{\odot}$$

$$P = 167 \text{ day}$$

$$e = 0.61$$

$$Z_{\star} = 0.06 Z_{\odot}$$

$$M_{\text{BH}} = 7.68 \pm 0.5 M_{\odot}$$

$$M_{\star} = 0.61 \pm 0.05 M_{\odot}$$

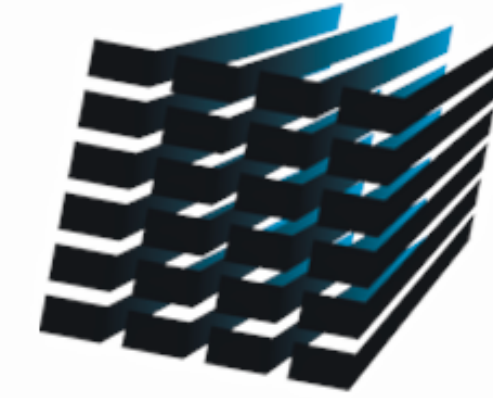
$$P = 2.11 \text{ day}$$

$$e = 0.10 \pm 0.14$$

$$Z_{\star} = 0.06 Z_{\odot}$$

GC models produce dynamically
formed BH - MS binaries —
indicates population of BHs in core

Eg Chatterjee+17, Kremer+18a,b



MUSE
multi unit spectroscopic explorer



Credit: ESO

3 low-mass BH candidates claimed in other surveys

J05210568

$$M_{\text{BH}} = 3.3^{+2.8}_{-0.7} M_{\odot}$$

$$M_{\text{LC}} = 3.2^{+1.0}_{-1.0} M_{\odot}$$

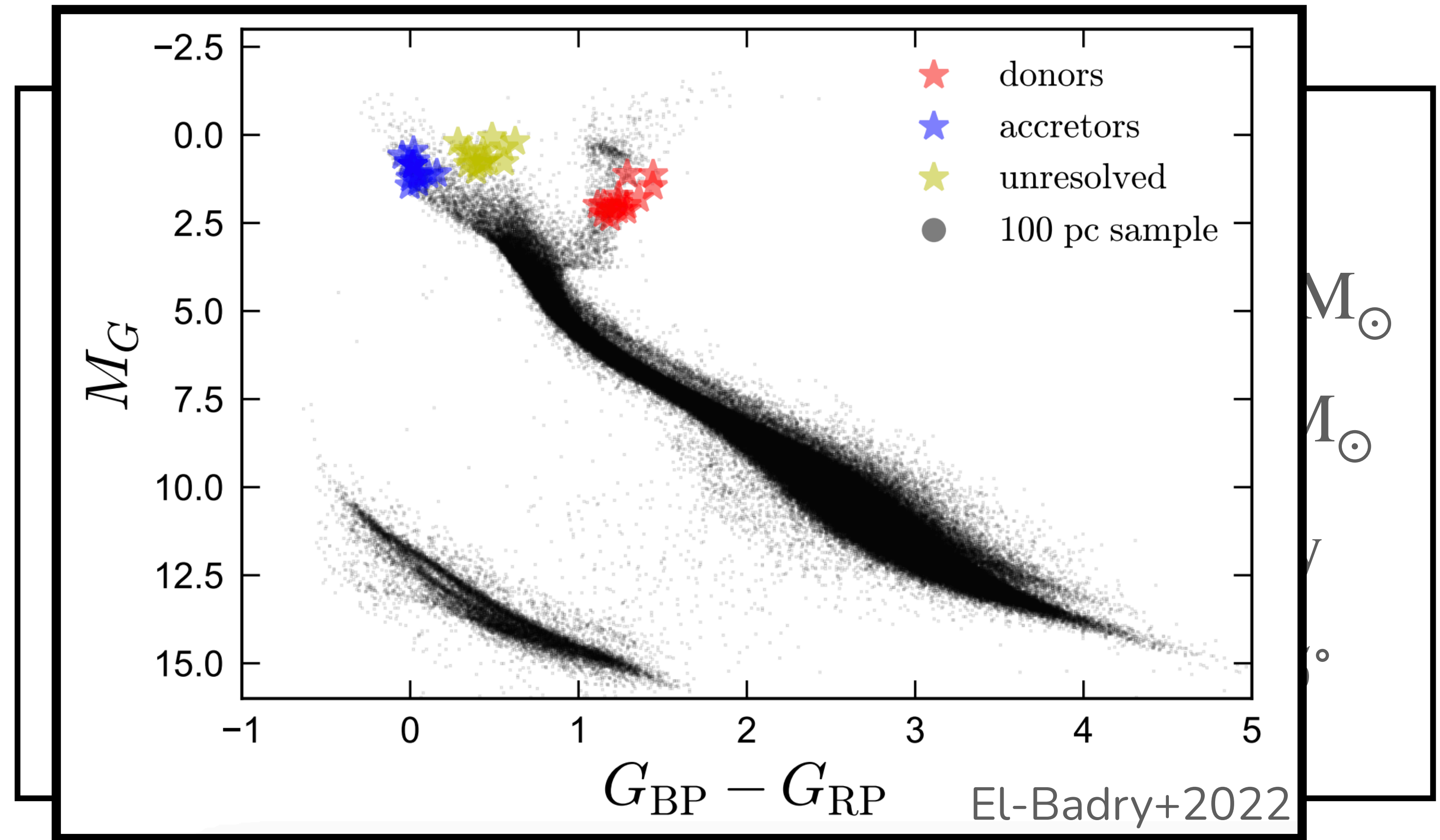
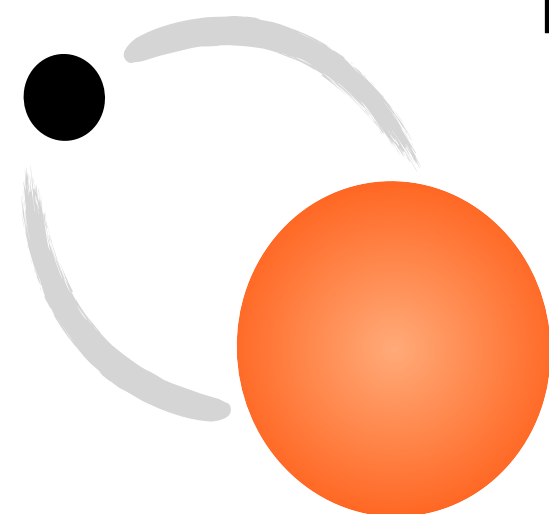
$$P_{\text{orb}} = 83.2 \text{ day}$$

$$e = 0, \quad \sin(i) = 0.97$$

Thompson+2019

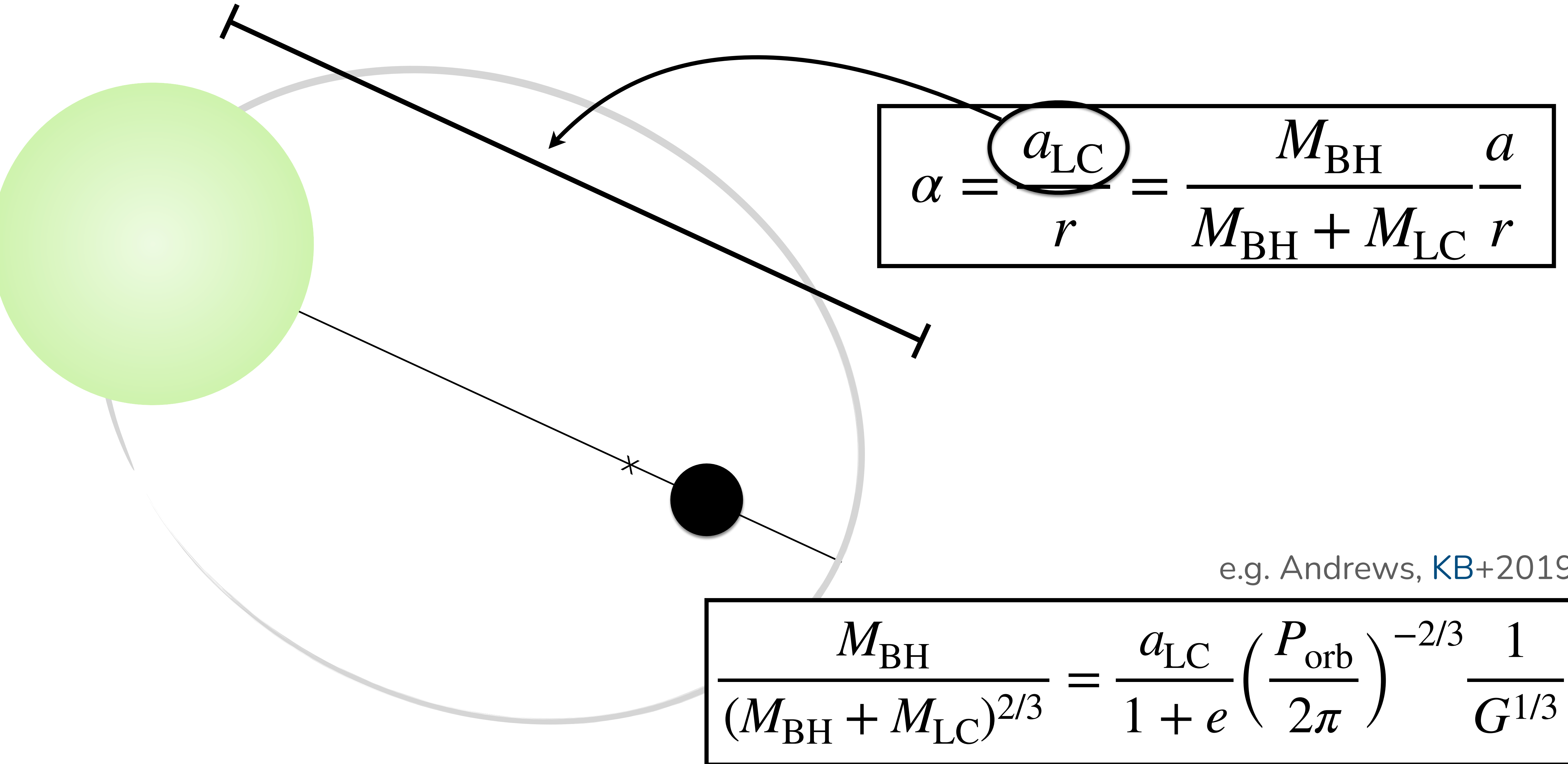
Jayasinghe+2021

Jayasinghe+2022



These turned out to be false positives

astrometric black hole binaries

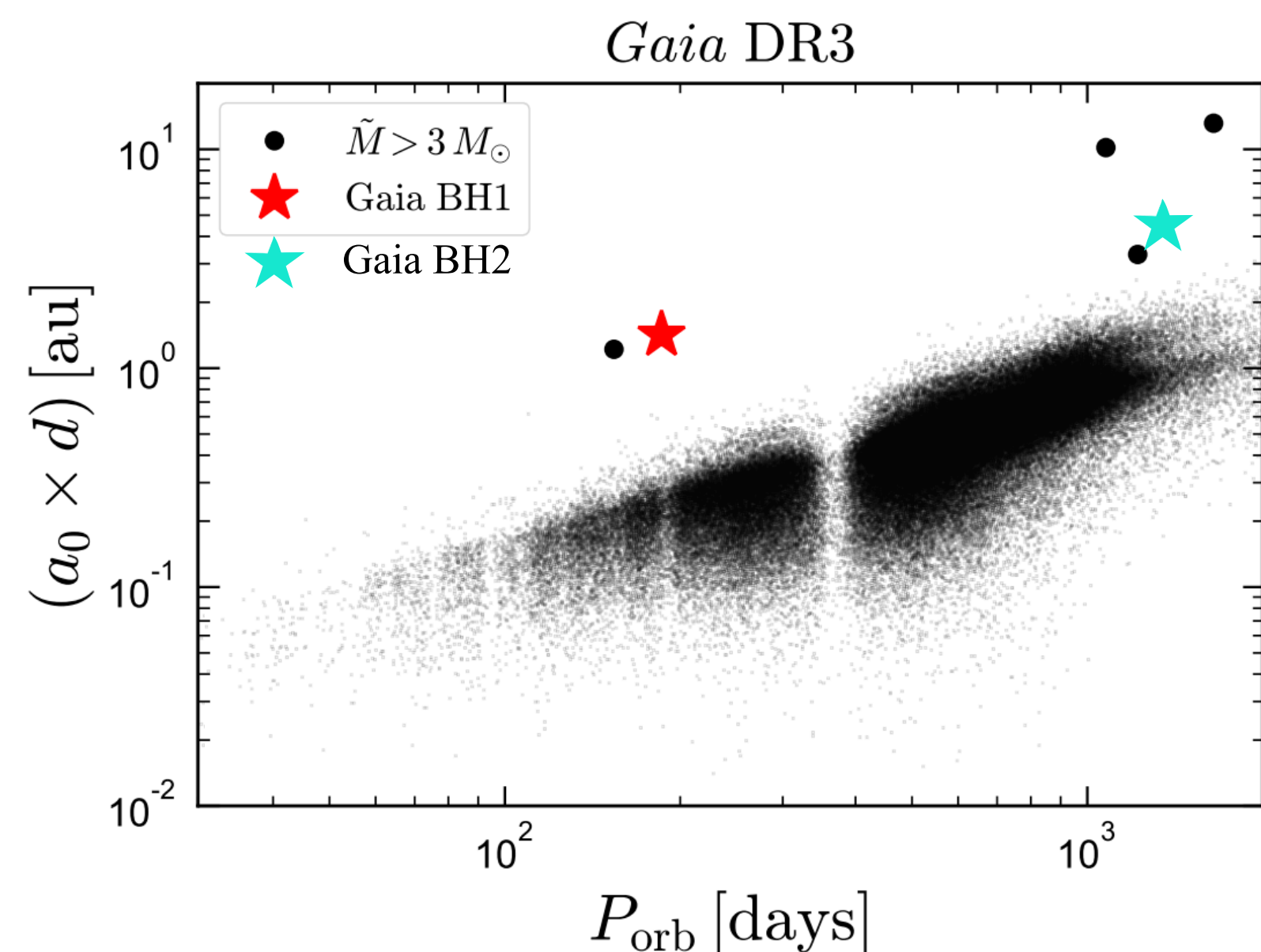


Gaia BH1/BH2

$M_{\text{BH1}} = 9.8 M_{\odot}$, $M_{\star} = 0.9 M_{\odot}$, $P = 185 \text{ day}$, $e = 0.45$, $Z_{\star} = 0.6 Z_{\odot}$, $D=460 \text{ pc}$

$M_{\text{BH2}} = 8.9 M_{\odot}$, $M_{\star} = 1.1 M_{\odot}$, $P = 1277 \text{ day}$, $e = 0.52$, $Z_{\star} = 0.6 Z_{\odot}$, $D=1.16 \text{ kpc}$

El-Badry+(incl KB)23a,b



an anomalously large
orbit for its period!

